

Supplemental Material

Table S1. General characteristics of the study participants in 3 metropolitan areas.

	Frequencies of moderate and vigorous PA (per week)			
	none	1 to 2	3 to 4	≥5
Number of people	27,937	11,033	8,238	11,907
Air pollution, mean (SD)				
PM10 (µg/m ³)	51.91 (5.35)	51.79 (5.39)	51.72 (5.49)	51.61 (5.58)
PM2.5 (µg/m ³)	26.50 (3.39)	26.35 (3.39)	26.32 (3.37)	26.25 (3.41)
Age, years, mean (SD)	52.1 (9.9)	49.3 (8.5)	50.6 (8.9)	52.8 (9.3)
Age, years, N (%)				
40-50	10,736 (38.43)	5,397 (48.92)	3,490 (42.36)	3,874 (32.54)
50-60	9,403 (33.66)	3,810 (34.53)	2,945 (35.75)	4,447 (37.35)
60-70	5,432 (19.44)	1,451 (13.15)	1,418 (17.21)	2,751 (23.10)
≥70	2,366 (8.47)	375 (3.40)	385 (4.67)	835 (7.01)
Body mass index, kg/m ² , mean (SD)	23.7 (3.1)	23.8 (3.0)	23.9 (3.0)	24.0 (2.8)
Body mass index, kg/m ² , N (%)				
<18.5	885 (3.17)	271 (2.46)	179 (2.17)	181 (1.52)
18.5-22.9	11,118 (39.80)	4,269 (38.69)	3,150 (38.24)	4,403 (36.98)
23.0-24.9	7,152 (25.60)	2,956 (26.79)	2,235 (27.13)	3,418 (28.71)
25.0-29.9	7,948 (28.45)	3,226 (29.24)	2,427 (29.46)	3,569 (29.97)
≥30.0	834 (2.99)	311 (2.82)	247 (3.00)	336 (2.82)
Sex, N (%)				
Men	11,515 (41.22)	6,033 (54.68)	4,386 (53.24)	6,162 (51.75)
Women	16,422 (58.78)	5,000 (45.32)	3,852 (46.76)	5,745 (48.25)
Household income, quartile, N (%)				
1 st (highest)	8,357 (29.91)	4,191 (37.99)	3,213 (39.00)	4,575 (38.42)
2 nd	8,284 (29.65)	3,206 (29.06)	2,334 (28.33)	3,308 (27.78)
3 rd	6,602 (23.63)	2,255 (20.44)	1,575 (19.12)	2,295 (19.27)
4 th (lowest)	4,694 (16.80)	1,381 (12.52)	1,116 (13.55)	1,729 (14.52)
Smoking, N (%)				
Never	19,302 (69.09)	6,369 (57.73)	4,898 (59.46)	7,435 (62.44)
Former	3,076 (11.01)	1,934 (17.53)	1,587 (19.26)	2,428 (20.39)
Current	5,559 (19.90)	2,730 (24.74)	1,753 (21.28)	2,044 (17.17)
Alcohol intake, times per week, mean (SD)	0.9 (1.5)	1.1 (1.4)	1.1 (1.4)	1.1 (1.6)
Alcohol intake, times per week, N (%)				
0	17,720 (63.43)	5,142 (46.61)	3,972 (48.22)	6,118 (51.38)
1-2	6,646 (23.79)	4,291 (38.89)	3,042 (36.93)	3,756 (31.54)
3-4	2,280 (8.16)	1,214 (11.00)	934 (11.34)	1,363 (11.45)

≥ 5	1,291 (4.62)	386 (3.50)	290 (3.52)	670 (5.63)
Systolic blood pressure, mmHg, mean (SD)	123.1 (15.79)	122.5 (14.9)	122.7 (14.9)	123.9 (15.2)
Fasting serum glucose, mg/dL, mean (SD)	99.4 (25.1)	98.3 (22.5)	99.1 (23.9)	99.9 (23.9)
Total cholesterol, mg/dL, mean (SD)	201.0 (37.1)	200.9 (35.7)	199.9 (36.3)	200.1 (36.3)
Charlson comorbidity index, mean (SD)	1.4 (1.4)	1.2 (1.3)	1.3 (1.4)	1.4 (1.5)
Charlson comorbidity index, N (%)				
0	8,113 (29.04)	3,631 (32.91)	2,507 (30.43)	3,422 (28.74)
1	9,372 (33.55)	3,813 (34.56)	2,813 (34.15)	3,883 (32.61)
≥ 2	10,452 (37.41)	3,589 (32.53)	2,918 (35.42)	4,602 (38.65)

N the number of people; *SD* standard deviation; *PA* physical activity

Table S2. Hazard ratios for cardiovascular disease risk according to frequencies of moderate to vigorous PA stratified by level of PM2.5.

		Frequencies of moderate to vigorous PA (per week)				<i>p</i> for trend	<i>p</i> for interaction
		none	1 to 2	3 to 4	≥5		
Number of people	Low/Moderate PM2.5 (<27.86µg/m3)	18,947	7,792	5,843	8,419		
	High PM2.5 (≥27.86µg/m3)	8,990	3,241	2,395	3,488		
Cardiovascular disease	Low/Moderate PM2.5 (<27.86µg/m3)						0.5
	Events, N (%)	337 (1.78)	85 (1.09)	83 (1.42)	118 (1.40)		
	Person-year	56,039	23,191	17,353	24,974		
	aHR (95% CI)	1.00 (reference)	0.73 (0.57-0.93)	0.86 (0.67-1.09)	0.73 (0.59-0.91)	0.004	
	High PM2.5 (≥27.86µg/m3)						
	Events, N (%)	161 (1.79)	46 (1.42)	32 (1.34)	41 (1.18)		
	Person-year	26,587	9,620	7,106	10,348		
	aHR (95% CI)	1.00 (reference)	0.91 (0.65-1.28)	0.80 (0.54-1.17)	0.62 (0.44-0.88)	0.004	
Coronary heart disease	Low/Moderate PM2.5 (<27.86µg/m3)						0.7
	Events, N (%)	150 (0.79)	49 (0.63)	48 (0.82)	59 (0.70)		
	Person-year	56,039	23,191	17,353	24,974		
	aHR (95% CI)	1.00 (reference)	0.89 (0.64-1.24)	1.04 (0.75-1.45)	0.79 (0.58-1.07)	0.2	
	High PM2.5 (≥27.86µg/m3)						
	Events, N (%)	69 (0.77)	24 (0.74)	15 (0.63)	22 (0.63)		
	Person-year	26,587	9,620	7,106	10,348		
	aHR (95% CI)	1.00 (reference)	1.00 (0.62-1.62)	0.81 (0.46-1.43)	0.73 (0.45-1.18)	0.2	
Stroke	Low/Moderate PM2.5 (<27.86µg/m3)						0.7
	Events, N (%)	193 (1.02)	38 (0.49)	35 (0.60)	59 (0.70)		
	Person-year	56,039	23,191	17,353	24,974		
	aHR (95% CI)	1.00 (reference)	0.61 (0.43-0.87)	0.68 (0.47-0.98)	0.68 (0.50-0.91)	0.003	
	High PM2.5 (≥27.86µg/m3)						
	Events, N (%)	92 (1.02)	23 (0.71)	17 (0.71)	19 (0.54)		
	Person-year	26,587	9,620	7,106	10,348		
	aHR (95% CI)	1.00 (reference)	0.86 (0.54-1.38)	0.78 (0.46-1.32)	0.53 (0.32-0.87)	0.012	

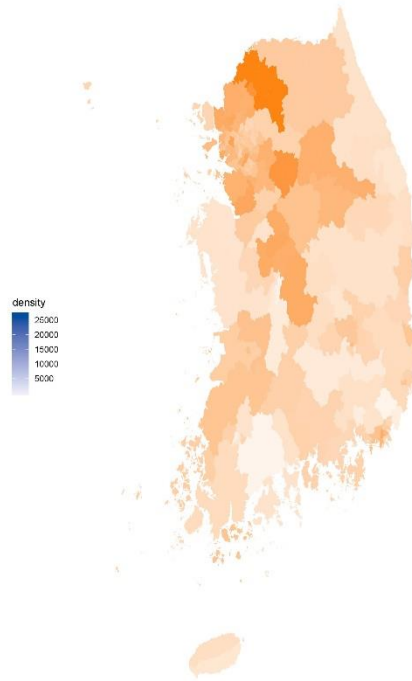
aHR (95 % CI) were calculated using Cox proportional hazards regression analysis after adjusting for location, age, sex, household income, body mass index, smoking, alcohol intake, systolic blood pressure, fasting serum glucose, total cholesterol, and Charlson comorbidity index.
PA physical activity; *aHR* adjusted hazard ratio; *CI* confidence interval.

Figure S1. a) Map of the distribution of population density (number of people/km²) of the cohort in 2010. b) Map of the distribution of air pollution (PM10) exposure (annual average concentrations in 2009 and 2010). c) Map of the distribution of air pollution (PM2.5) exposure in the 3 metropolitan cities (annual average concentrations in 2009 and 2010).

a)



b)



c)

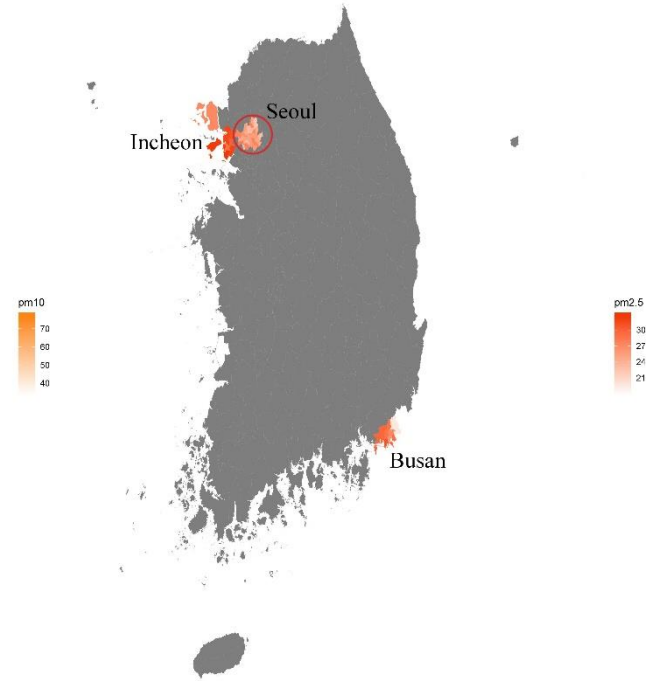


Figure S2. Flow diagram of selection of study participants in 3 metropolitan cities.

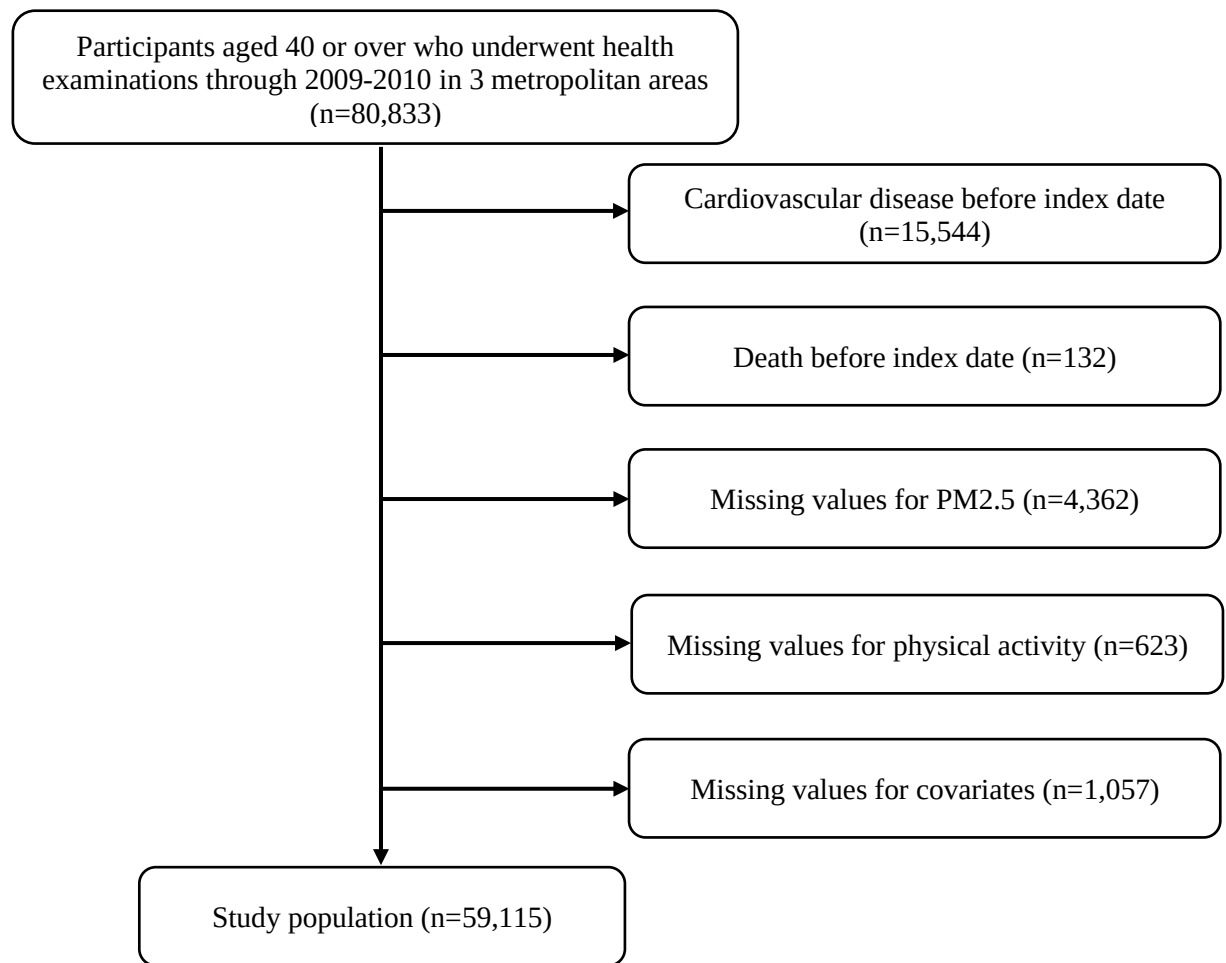
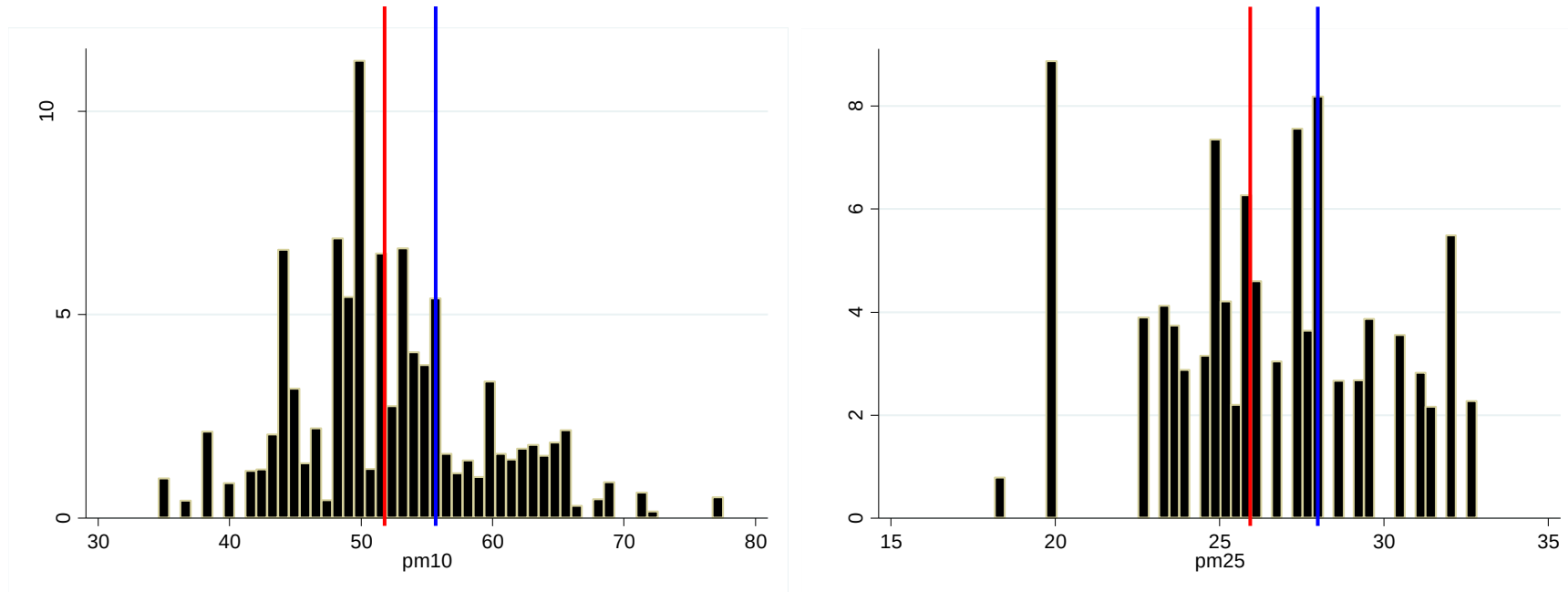


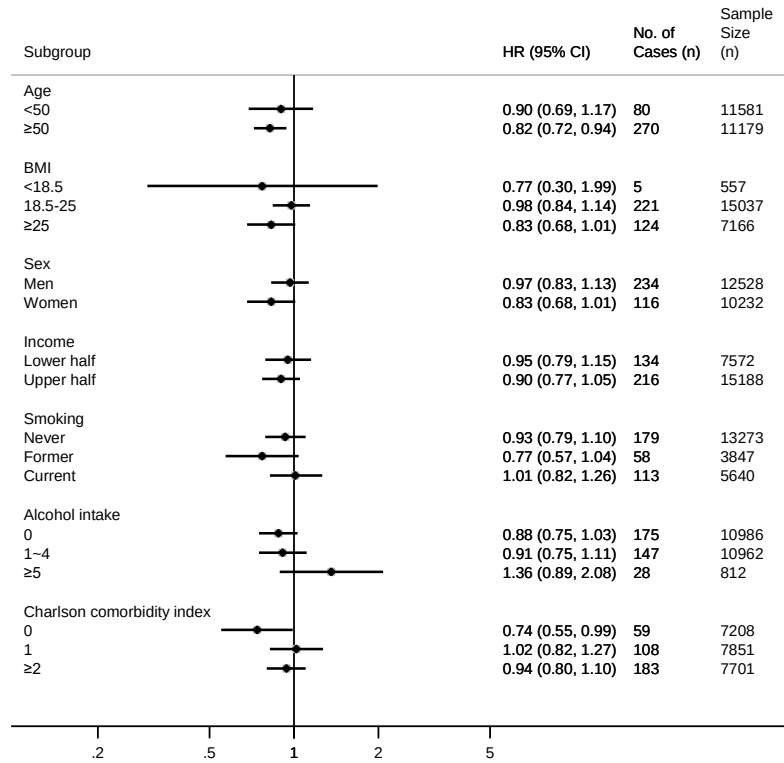
Figure S3. Graph of the frequency distribution of PM10 and PM2.5 exposure (annual average concentrations in 2009 and 2010) among the study participants.



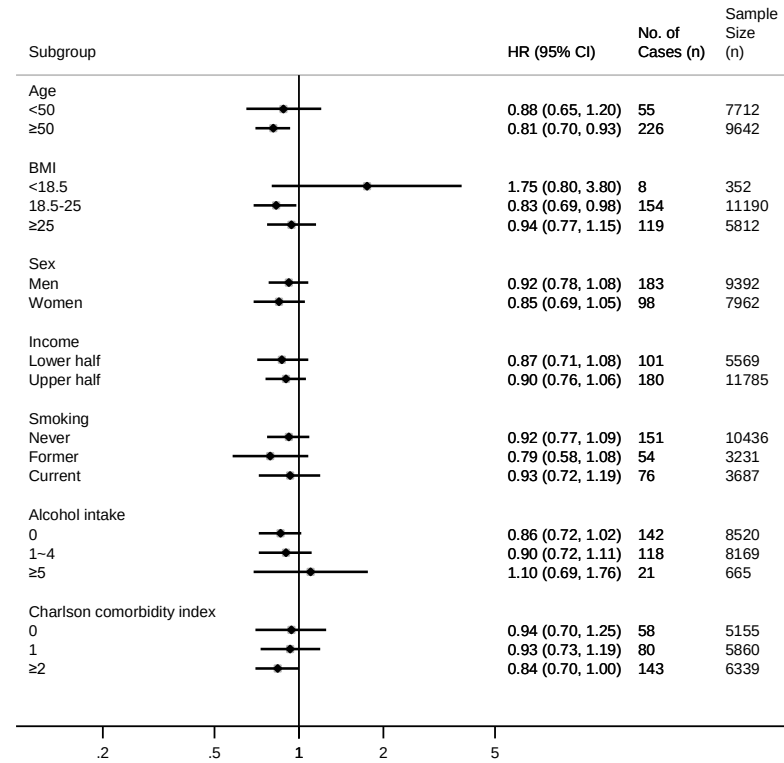
The 50th percentile of exposure range was 51.54 $\mu\text{g}/\text{m}^3$ for PM10 and 26.18 $\mu\text{g}/\text{m}^3$ for PM2.5, which is represented as the red line. The 70th percentile of exposure range was 55.13 $\mu\text{g}/\text{m}^3$ for PM10 and 27.86 $\mu\text{g}/\text{m}^3$ for PM2.5, which is represented as the blue line.

Figure S4. aHR (95 % CI) of CVD by subgroups according to the level of PM10. All aHRs (95 % CI) shown above are adjusted for age, sex, household income, body mass index, smoking, alcohol intake, systolic blood pressure, fasting serum glucose, total cholesterol, and Charlson comorbidity index.

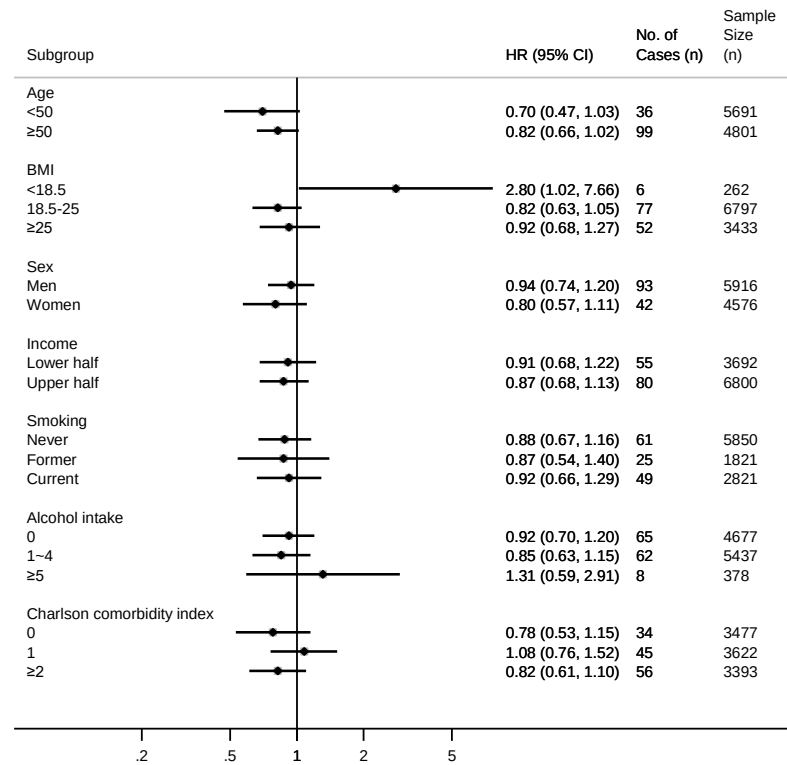
A. Subjects with moderate to vigorous 1 to 2 times/week PA and low PM10



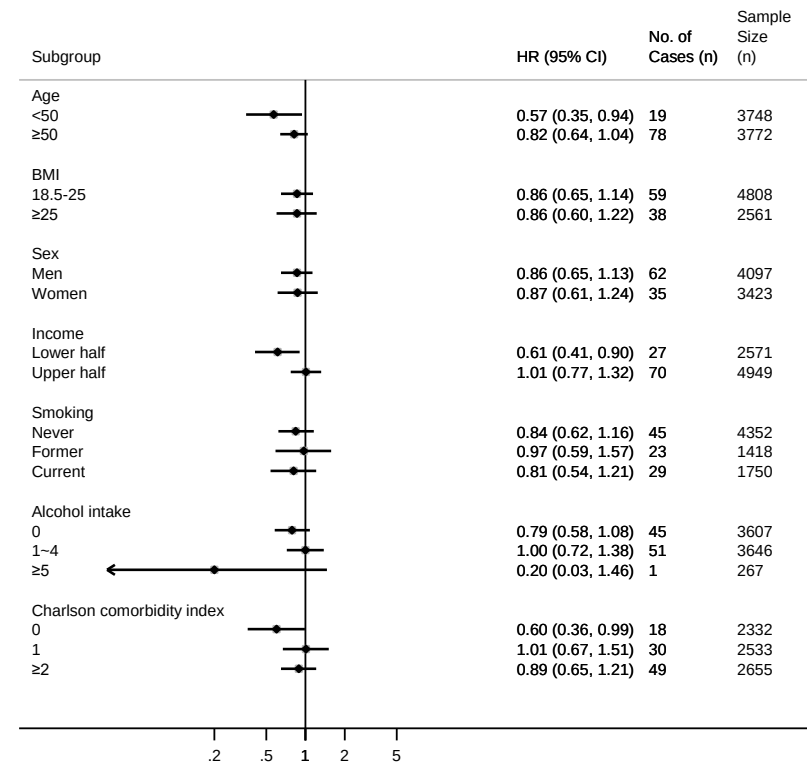
B. Subjects with moderate to vigorous PA 3 to 4 times/week and low PM10



C. Subjects with moderate to vigorous 1 to 2 times/week PA and high PM10



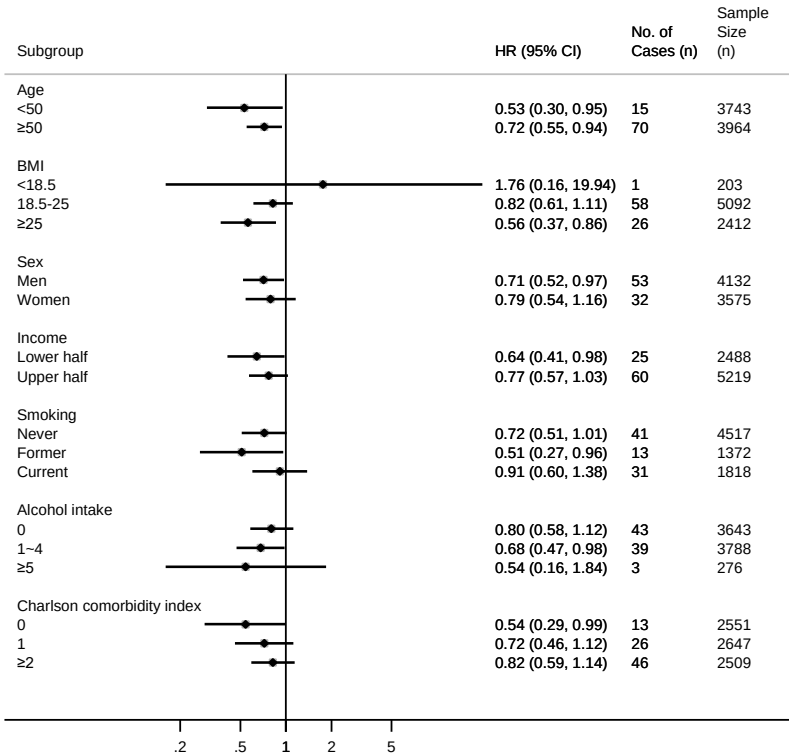
D. Subjects with moderate to vigorous 3 to 4 times/week PA and high PM10



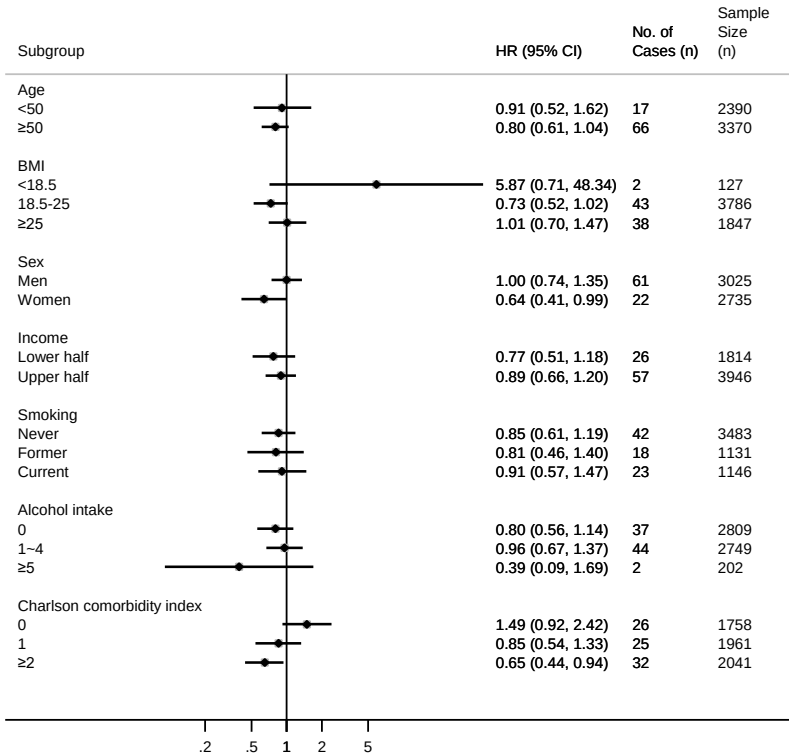
Individuals with no moderate to vigorous PA are the reference group for all subgroup analyses. *CVD* cardiovascular disease; *PM* particulate matter; *PA* physical activity; *aHR* adjusted hazard ratio; *CI* confidence interval.

Figure S5. aHR (95 % CI) of CVD by subgroups according to the level of PM2.5. All aHRs (95 % CI) shown above are adjusted for age, sex, household income, body mass index, smoking, alcohol intake, systolic blood pressure, fasting serum glucose, total cholesterol, and Charlson comorbidity index.

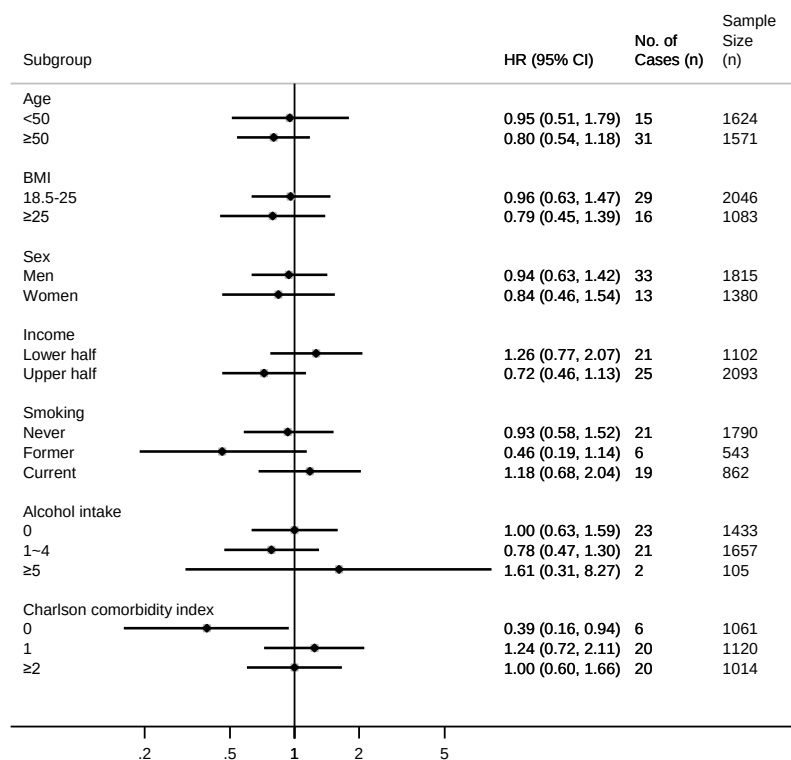
A. Subjects with moderate to vigorous 1 to 2 times/week PA and low PM2.5



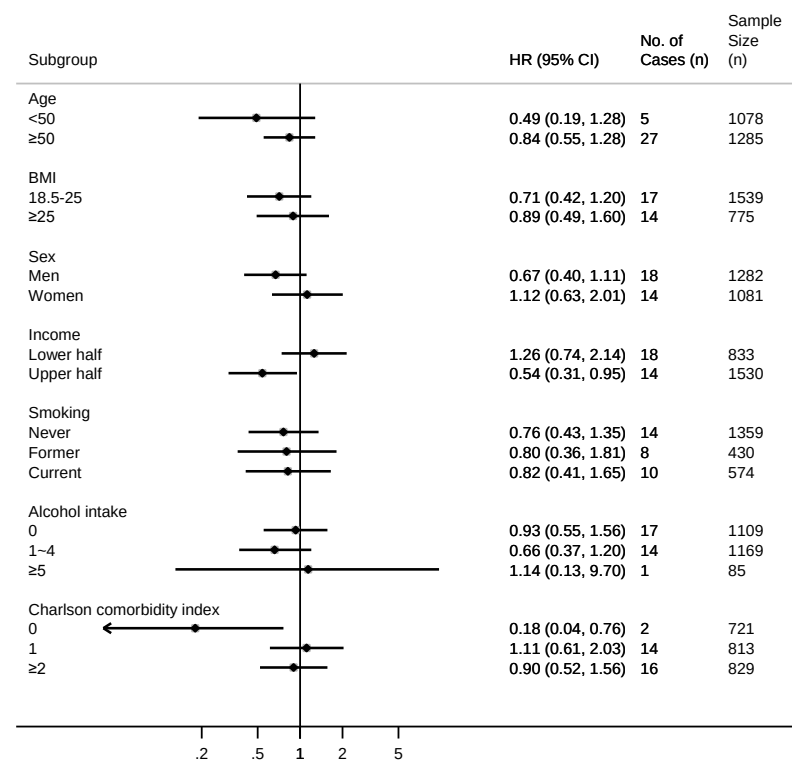
B. Subjects with moderate to vigorous 3 to 4 times/week PA and low PM2.5



C. Subjects with moderate to vigorous 1 to 2 times/week PA and high PM2.5



D. Subjects with moderate to vigorous 3 to 4 times/week PA and high PM2.5



Individuals with no moderate to vigorous PA are the reference group for all subgroup analyses. *CVD* cardiovascular disease; *PM* particulate matter; *PA* physical activity; *aHR* adjusted hazard ratio; *CI* confidence interval.